

RASWANTH MURUGAN

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OBJECTIVE

Research scientist with four years of experience in reinforcement learning, LLM post-training, and generative modeling. At Caltech, built the LLM fine-tuning pipeline for in-silico enzyme design, covering both SFT and GRPO, and scaled it across the Polaris HPC cluster; equal-contribution author on the resulting MLSB 2025 @ NeurIPS 2025 paper. Earlier, built conditional generative models for molecular design at CUNY/Weill Cornell (AI4Science @ ICML 2024) and trained offline RL agents for autonomous underwater navigation at IISc (AAMAS 2023).

EDUCATION

Indian Institute of Technology, Palakkad (Kerala)

Bachelor of Technology, Computer Science and Engineering

July 2018 – July 2022

CGPA: 8.66/10.00

TECHNICAL SKILLS

RL & Post-Training: GRPO, SFT, self-distillation, reward design & reward shaping, multi-objective / multi-fidelity reward composition, offline RL, online RL, Decision & Trajectory Transformers, policy evaluation, ablation design

Generative Models & LLMs: latent diffusion, VAEs, contrastive (CLIP-style) cross-modal alignment, conditional generation, sequence models, protein language models

Engineering & Systems: Python, PyTorch, `torch.distributed`, multi-node HPC (PBS Pro / ALCF Polaris), end-to-end pipeline ownership, simulation & evaluation harnesses, Hugging Face, Docker, Linux, Git; daily power user of Claude Code for research engineering

WORK EXPERIENCE

AI+Science Lab – Caltech & Argonne National Laboratory

AI-driven D-Handed Enzyme Design

Guides: [Anima Anandkumar](#), [Arvind Ramanathan](#), [Frances Arnold \(Nobel Prize 2018\)](#)

Apr 2025 – Present

Research Scientist

Frances Arnold pioneered directed evolution – bringing evolution into the lab (in vitro) to design enzymes – which won her the 2018 Nobel Prize. While transformative, the process remains slow and often struggles to design enzymes with the desired function and efficiency. This project takes the next step by moving directed evolution in silico using AI and LLMs; as a demonstration, we redesign the TrpB enzyme to catalyse a non-natural reaction entirely on the computer.

- Designed a multi-fidelity GRPO reward function for in-silico directed evolution: with exact activation-energy QM out of compute budget, composed cheaper biophysical surrogates (semi-empirical QM, structure prediction, docking) into separate reward terms – protein stability, complex stability, substrate binding, and reaction coordinates – each chosen to capture a distinct stage of enzyme catalysis.

Note: Main repository access will be provided on request as work is still in progress

- Owned the full in-silico directed evolution pipeline end-to-end (PyTorch / Python): wrote the SFT codebase from scratch, designed and implemented the GRPO trainer, built in-silico filtering layers, and assembled the MD validation + analysis workflow – learning the computational chemistry stack from the ground up (AutoDock Vina, g-xTB semi-empirical QM, FoldX stability, MD simulation, PyMOL-based structural analysis).
- Scaled GRPO and SFT training on Polaris HPC cluster (ALCF): PBS Pro job scripts, `torch.distributed`; batch-32 GRPO steps reduced from 4.3 hr → 14 min ($\approx 18\times$); SFT backbone generation per Theozyme reduced from 9 hr → <2 hr [\[Enzyme Design GitHub\]](#).
- Co-first author, MLSB @ NeurIPS 2025: drove all training, evaluation, and ablation experiments on GenSLM-25M / TrpB; demonstrated that self-distillation SFT lifts catalytic Lys conservation 95% → 99.6% and – more importantly – yields emergent gains on properties never directly optimized: Rosetta stability mean shifts 2.49 → 10.55 ($\approx 4\times$) and PLP docking score 4.75 → 6.03 (+27%), validating the recipe as a general protein-design framework without explicit functional reward shaping [\[SFT GitHub\]](#).

Xie Lab – City University of New York & Weill Cornell Medicine

Inverse Molecular Design to Modulate Single-Cell Transcriptomics

Guide: [Lei Xie](#) | J-1 Fully Sponsored Visiting Scholar

Sep 2023 – Apr 2025

Research Scientist, New York

Given a description of a diseased state, can generative models design small molecules that move it toward a healthy one? Inspired by text-to-image models, we built a generative AI pipeline that uses gene-expression profiles of healthy and diseased states as input to synthesize candidate drug-like molecules tailored to the phenotype.

- Pioneered MolGene-E for phenotype-driven drug discovery (ICML 2024 AI for Science Workshop): bypassed traditional structure-based docking by mapping single-cell transcriptomic profiles directly to novel small molecules; pipeline = denoising VAE over L1000toRNAseq (MCF7; 3,116 high-variance genes) + CLIP-style contrastive alignment of gene $\leftrightarrow$ SELFIES embeddings + MLP prior + frozen SELFIES VAE decoder.
- Achieved state-of-the-art zero-shot generation: validity 1.00, novelty 1.00, QED 0.69 on CRISPR knock-out profiles (1,218 L1000 signatures, 44 transcription factors); outperformed MolGAN (validity 0.29) and GxVAEs (validity 0.27); statistically significant similarity gains across MACCS, Fraggle, and ErG fingerprints (Mann-Whitney U , $p < 0.05$); generalized bulk $\rightarrow$ single-cell on SciPlex-3 (188 compounds) without fine-tuning.
- Independently engineered a DALL-E-inspired latent diffusion language model for SELFIES generation: compressed variable-length SELFIES via an encoder-decoder into a fixed-dimensional continuous latent space, applied gene-expression-conditioned latent diffusion, decoded back to molecular structures; full implementation at [\[LD4LG GitHub\]](#).
- Conducted multi-cell-line scaling analysis uncovering critical data sparsity and noise limitations in zero-shot biological mapping; demonstrated that standard LLM/diffusion architectures must be augmented with first-principle biophysical inductive biases for cross-cellular generalization – directly informing future AI-for-Science methodology.

Samsung R&D Institute

On-device topology change detection using UWB radar sensor data
Team: UWB Intelligence

Sep 2022 – Sep 2023

ML Engineer, Bengaluru

UWB sensors excel at precise localization, enabling solutions like low-light focus enhancement for Galaxy S-series devices. Their capabilities can also be leveraged to develop navigation aids for visually impaired users. This project explored the application of ML on UWB radar sensors for depth estimation and topology change detection.

- Engineered an end-to-end 2D-CNN pipeline on Ultra-Wideband (UWB) sensor data: 2D Channel Impulse Response (CIR) radargrams (complex-valued distance $\times$ time-of-flight matrices) preprocessed via signal-processing denoising chain, then fed to CNN for joint object classification and distance regression.
- Designed a closed-loop active learning system to bootstrap a UWB-CIR dataset from scratch (no public benchmark existed): iterative cycles of in-house data collection $\rightarrow$ training $\rightarrow$ evaluation (precision / recall / ROC) $\rightarrow$ targeted re-collection of uncertain samples.
- Spearheaded product ideation and proof-of-concept demonstrations leveraging UWB for zero-light spatial awareness: night-mode camera enhancements for lower-tier Galaxy S-series devices and collision-detection accessibility features for visually-impaired users; owned the Python ML pipeline end-to-end and handed off trained models to the hardware team for on-device deployment.

Indian Institute of Science (IISc)

Intelligent Onboard Routing in Stochastic Environments using Transformers
Guide: [Deepak Subramani](#) (CDS, IISc)

Jan 2022 – Jan 2023

Research Fellow, Bengaluru

Time-optimal path planning in dynamic ocean velocity fields requires addressing sparse observations and computational constraints. This project explored an end-to-end offline RL approach using transformers to reframe planning as a sequence prediction task, enabling implicit learning of both state estimation and planning for efficient and accurate trajectory optimization.

- Prototyped data assimilation using Message Passing Networks (PGMs and MPNNs) to estimate the joint distribution $P(\text{Discretized Ocean State} \mid \text{Prior Belief, Observations, Sensor Noise})$; validated on simple joint Gaussian testbeds before scaling to ocean velocity fields.
- Implemented and integrated a Trajectory Transformer pipeline (coupled with beam search) directly into the main AUV codebase; trained and evaluated end-to-end on generated simulation data.
- Conducted comparative analysis between Trajectory and Decision Transformers; identified the key architectural insight that prepending a destination token to the Decision Transformer's sequence significantly improved multi-target generalization.
- Validated across 8 Double Gyre flow test cases; second author on AAMAS 2023 publication on intelligent onboard routing in stochastic dynamic environments. [\[Decision Transformer GitHub\]](#).

PUBLICATIONS

- Amin Tavakoli*, **Raswanth Murugan***, Ozan Gokdemir, Arvind Ramanathan, Frances H Arnold, Anima Anandkumar. "Self Distillation Fine-Tuning of Protein Language Models Improves Versatility in Protein Design". Machine Learning in Structural Biology (MLSB) @ NeurIPS 2025, Copenhagen & San Diego. [\[arxiv\]](#)
- Rahul Ohlan, **Raswanth Murugan**, Li Xie, Mohammadsadeq Mottaqi, Shuo Zhang, and Lei Xie. "MolGene-E: Inverse Molecular Design to Modulate Single Cell Transcriptomics". ICML 2024 AI4Science Workshop, Austria. [\[openreview\]](#)

- Rohit Chowdhury, **Raswanth Murugan**, and Deepak Subramani. "Intelligent Onboard Routing in Stochastic Dynamic Environments using Transformers". AAMAS 2023, London. [\[acm\]](#)

SCHOLASTIC ACHIEVEMENTS

- **PhD Admit** – A Fully Sponsored PhD offer from [Xie Lab](#) was received but turned down.
- **Summer Research Fellowship 2021** – Indian Academy of Sciences (2.5% acceptance rate, ~120 nationwide awardees).
- **Gold Medal** – Inter-IIT Tech Meet 2018–2019, IIT Bombay – Autonomous Pesticide Spraying Robot (Tata Centre for Technology and Design).
- **95th percentile** – JEE Advanced (0.23 million candidates).
- **99.4th percentile** – JEE Mains (1.3 million candidates).
- **State Rank 10** – National Talent Search Examination (NTSE), Government of India, 2015.

PROJECTS

Project meet.me

Aug 2021 – Dec 2021

UX Team Lead | IIT Palakkad | Guide: Mr. Ramaswamy Krishnan-Chittur (Microsoft)

Collaborative meeting app with real-time screen sharing and whiteboard, built by a 27-developer team. [\[GitHub\]](#)

TimeSync

Sep 2020

HackMIT

Accountability partner app to combat screen addiction; built with Flutter, Firebase, OneSignal. [\[GitHub\]](#)

Autonomous Pesticide Spraying Robot

Sep 2018 – Dec 2018

TCTD Innovation Challenge | Inter-IIT Tech Meet, IIT Bombay

Gold medal-winning autonomous robot for small-scale Indian farmers. [\[Details\]](#)

POSITIONS OF RESPONSIBILITY

- **Mathematics Tutoring** - Preply Sep 2025 – Present
Tutoring mathematics from grade 1 to Masters level; students from China, Australia(USQ), Germany, and USA (UC Davis, UC Irvine).
- **Member, People Task Force** – Samsung R&D Institute Bengaluru Dec 2022 – Sep 2023
Selected 1 of 6 from a 150-person Advanced Research Group; organised cultural events; reported to VP of Samsung R&D India.
- **Editor, The Fleet Street & Interviewer, Alumni Cell** – IIT Palakkad Jan 2021 – Dec 2021
Researched, wrote, and published student media; hosted fireside chats with faculty and alumni during COVID.
- **Student Mentor** – IIT Palakkad Jul 2020 – Dec 2020
Guided 10 freshmen on academic, holistic, and overall development.